

Opportunistic Target Selection

A tiny online target choice makes drift-prone black-box attacks behave like targeted attacks

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Core Idea

Class drift wastes black-box queries. OTS spends a few untargeted iterations discovering where the attack is already drifting, then commits to that class.

$$t = \arg \max_{k \neq y} P(k \mid x'_T) \quad x'_{i+1} \leftarrow A_{\text{targeted}}(x'_i, t)$$

+27 pp
success

SimBA on ResNet-50

43%
fewer
iterations

censored mean,
ResNet-50

4.5k
runs

standard ImageNet
benchmark

Why Untargeted Attacks Drift

Probability and CE attacks optimize only the true-class term:

$$\mathcal{L}_{\text{untarg}} = P(y \mid x') \quad \text{or} \quad \log P(y \mid x').$$

They lower confidence in y without deciding which non-true class should take over.

- ▶ The perturbation moves through several transient class basins.
- ▶ Targeted attacks remove drift but need a target class before the attack starts.
- ▶ Margin loss fixes this dynamically, but many attacks do not use it.

Algorithm

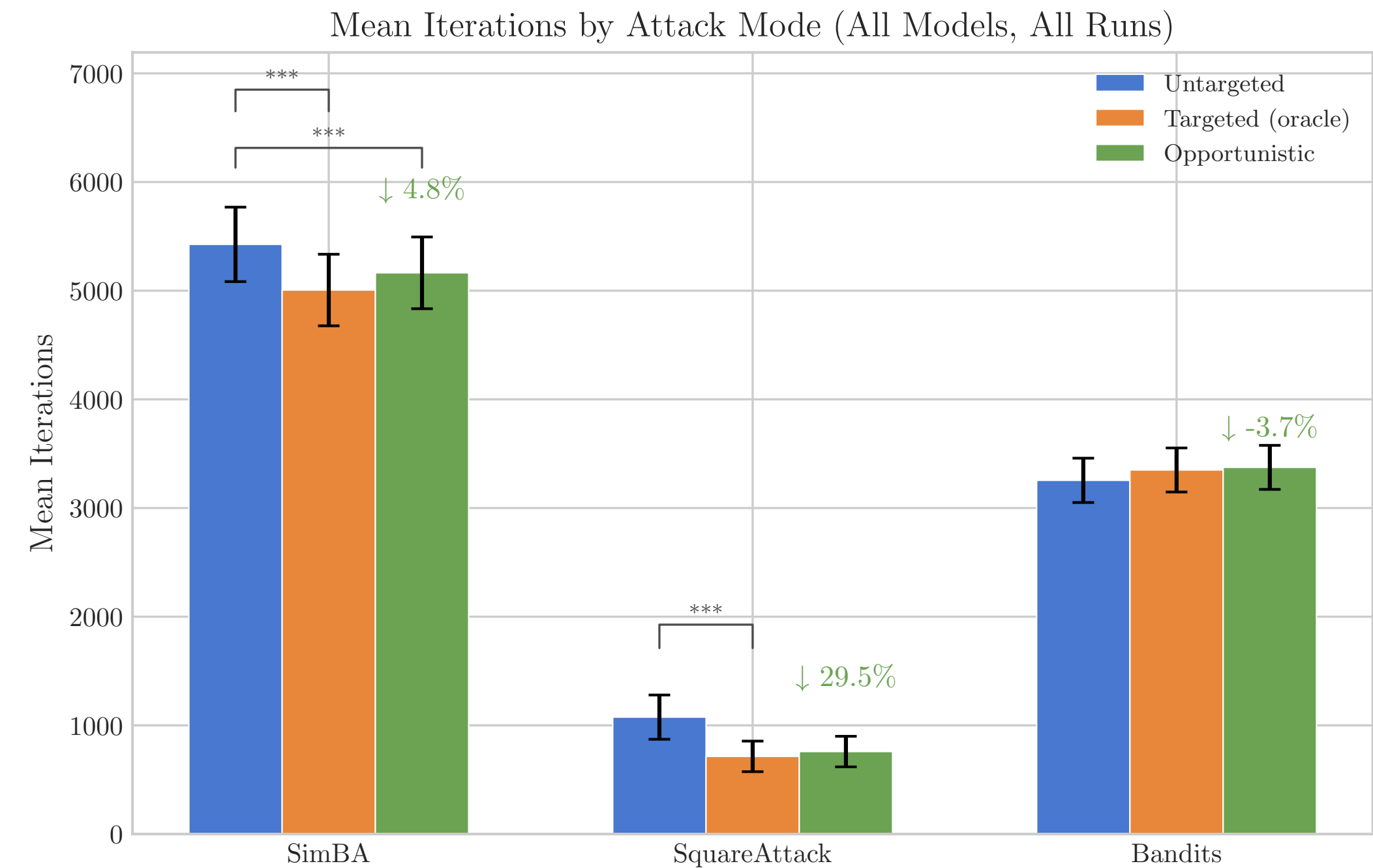
1. **Explore.** Run the original attack in untargeted mode for T iterations.
2. **Select.** Choose the leading non-true class under the current perturbed image.
3. **Exploit.** Keep the same perturbation mechanism, but switch to the targeted objective.

Setting	Default	Rationale
Budget	10K iters.	equal-query comparison
Perturbation	$\epsilon = 8/255$	ImageNet L_∞
Switch	$T = 5$ to 500	insensitive in ablations
Target	current top non-true	no a priori class needed

Interpretation

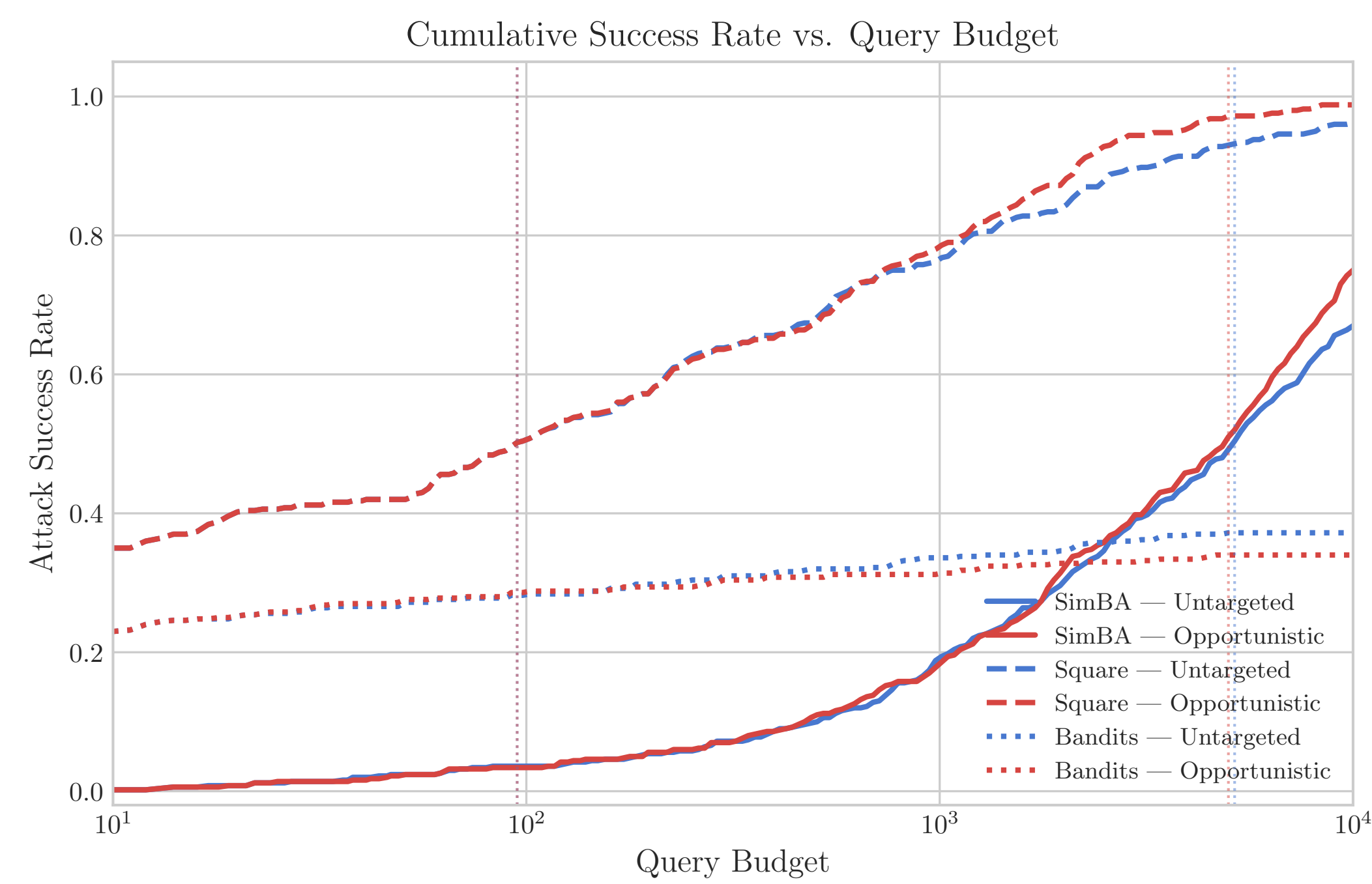
OTS is a **margin-loss surrogate**: after lock-in, it approximates the competitor term $\max_{k \neq y} f_k(x')$ with one fixed viable class t . It helps exactly when the attack lacks this competitor tracking.

Main Result: OTS Tracks the Oracle



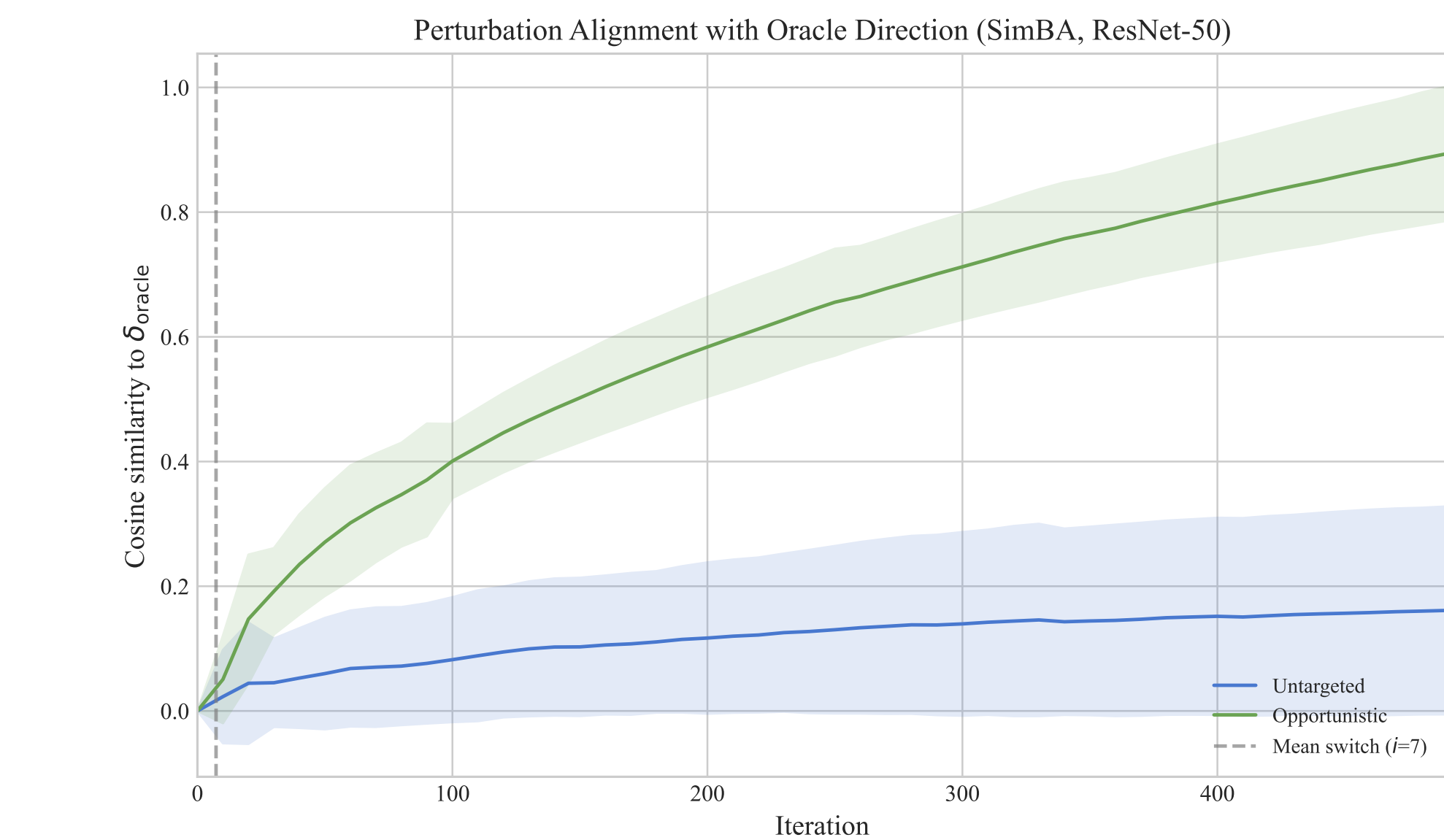
Attack	Untargeted	OTS	Oracle
SimBA	67%	75%	76%
Square Attack CE	96%	99%	99%
Bandits	37%	34%	35%

Query Budget View



The separation is clearest for SimBA and Square Attack with CE loss. Bandits remains flat because gradient estimation already gives directional information.

Alignment Mechanism



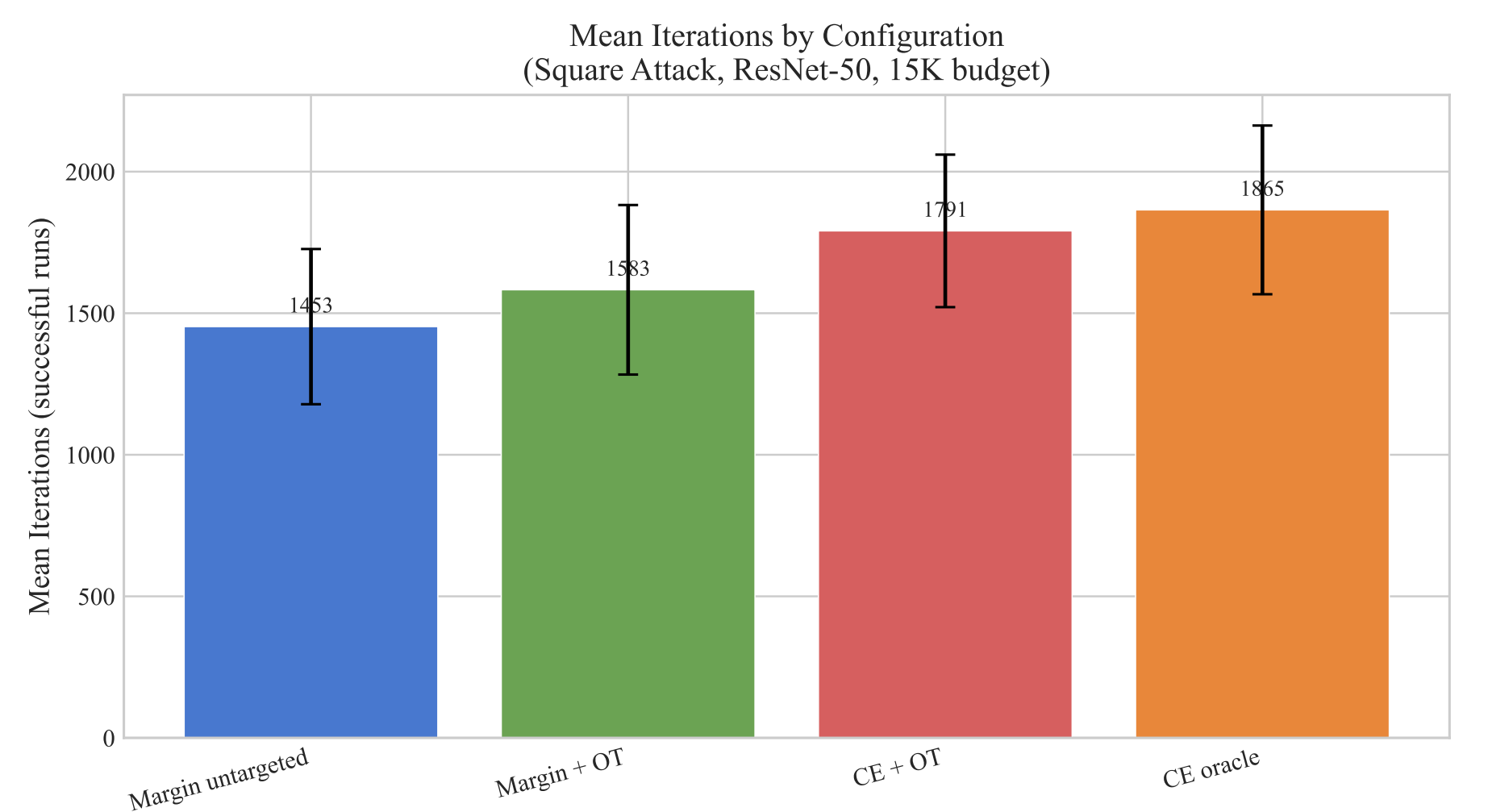
OTS does not merely select a class label. It redirects the perturbation toward the oracle targeted direction: terminal cosine similarity rises from 0.174 to 0.865 on SimBA/ResNet-50.

Where OTS Helps

Helps	Does not help
SimBA probability minimization	Bandits gradient estimation
Square Attack with CE loss	Square Attack margin loss
Medium-difficulty standard models	Robust models with early-success/fail split

- ▶ ResNet-50 is the strongest case: SimBA rises from 43% to 70% success.
- ▶ ViT-B/16 also benefits, so the effect is about attack difficulty rather than CNN depth.
- ▶ Easy models show floor effects because nearly every method already succeeds.

Margin-Loss Ablation



With CE loss, OTS restores much of the lost directionality. With margin loss, OTS is redundant.

Target Selection Matters

- ▶ Random targets are catastrophic: directional commitment alone is not enough.
- ▶ Clean-image argmax nearly matches OTS for SimBA.
- ▶ Short exploration is useful for Square Attack because patch updates disturb the class ranking more strongly.



Practical Rule

Use OTS as the default wrapper for score-based random-search attacks that optimize probability or CE objectives. Skip it for attacks already using margin loss or strong gradient-estimation priors.

Paper: arxiv.org/abs/2605.25663
Code: [Tardieu/opportunistic-target-selection](https://github.com/florent-tardieu/opportunistic-target-selection)